INCH-POUND

A-A-54859
21 May 1993

COMMERCIAL ITEM DESCRIPTION

TUBE, EXTENSION, INJECTION SET

The General Services Administration has authorized the use of this Commercial Item Description as a replacement for Military Specifications MIL-T-36673B and MIL-T-36192A(4) which are cancelled.

This Commercial Item Description covers three types of sterile, disposable, plastic, intravenous extension sets, suitable for the administration of intravenous medications and for the introduction of diagnostic dyes into the circulatory system.

Salient characteristics:

General. The units shall be disposable extension sets suitable for the administration of intravenous medications and for the introduction of diagnostic dyes into the circulatory system.

Material. All rubber and plastic components shall be sterile, free from pyrogens, compatible, safe (non-toxic), and shall be in accordance with the Pharmacopeia of the United States (USP) for Class VI plastics and for transfusion and infusion assemblies. The transparent PVC tubing shall be flexible and pliable, but be sufficiently rigid to prevent collapse during normal use. The tubing shall be free from all liquid additive substances which oxidize, exude or sweat-out onto the surface of the tubing.

The adapters and clamps shall be fabricated from a rigid, dimensionally stable plastic.

AMSC N/A FSC 6515

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Style, design and dimensions. The dimensions for the tubing of the three types of sets shall be as follows (dimensions are in inches):

TYPE	Overall Length	Internal Diameter	External Diameter
I	30 ± 4	0.100 ± .003	0.138 ± .005
II	120 ± 5	0.120 ± .003	0.170 ± .005
III	30 ± 4	0.100 ± .003	0.138 ± .005
IV	20 ± 3	0.100 ± .005	0.138 ± .005
V	30 ± 4	0.100 ± .005	0.138 ± .005

The extension sets shall have permanently attached female Luer adapters at one end and male slip-fit Luer adapters at the other end. The adapters shall each be fitted with a tight but readily removable end protector which shall maintain the sterility of the fluid path and adapters after sterilization. The Types I and II extension sets shall have a regulating roller clamp or a slide type shut-off clamp present to control the flow rate from complete shut-off to full flow. The extension sets shall be supplied with the clamps positioned on the tubing in the open position. The Type III extension set shall have two Y injection sites present 6 and 26 inches from the distal luer. A slotted slide clamp shall be present between the female Luer adapter and the Y injection site and a slide or regulating roller clamp between the two Y injection sites. The open end of the Y shall face the female adapter. The Types IV and V extension sets do not require a flow regulator.

Description of Components. The male slip-fit adapter shall be approximately 1-1/2 inches in overall length. The female adapter shall be 3/4 inch to 1-3/4 inches in overall length. The tapers shall be in accordance with ISO 594/1. The adapters shall not separate or looser from the plastic tubing.

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The flow regulator shall be a roller or slotted slide-type clamp and shall be permanently attached to the plastic tubing. The roller of the roller clamp shall not be removable. Slide clamps shall be rigid, smooth and free from burrs. It shall be moveable, and shall be capable of one-handed operation. The regulator shall be capable of accomplishing a complete quick shut-off, with no falling drop apparent in 5 minutes. The regulator shall be capable of maintaining a constant flow rate of liquid at any point between complete shut-off and maximum flow.

Each injection site shall be a rigid, transparent "Y" with a permanently attached puncturable, self-sealing rubber diaphragm.

Performance and Construction Requirements.

Flow rate. Flow rate for Types I, III, IV and V sets shall be 100 ml of water in no more than 120 seconds when connected to a regular administration set. The flow rate for the Type II set shall be 1 liter in less than 16 minutes when connected to a 1 liter bottle of water.

Leakage and separation of set. The adapter shall be permanently attached to the plastic tubing such that with the opening-blocked, it shall not leak or separate when subjected to an internal hydrostatic pressure of 45 psig for 5 minutes.

Deformation of tubing. While held in the same position on the tubing, the flow regulator shall be operated from its fully open position to complete shut-off ten times. The tubing shall not be cut, pierced, or damaged in any manner which would cause leakage or restriction of the flow of solution.

Diaphragm reseal test. The rubber diaphragm shall reseal without a falling drop within 10 seconds after puncture with a dry nonlubricated 18 gage needle when subjected to an internal (hydrostatic) pressure of 4 psig for 60 seconds.

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Security of roller in clamp body. When the flow regulator is of the roller type, the roller shall not be removed from the roller clamp under a 3.65 kilogram load when tested as follows:

The roller clamp shall be removed from the set together with a piece of tubing approximately one inch longer than the clamp body. The tubing shall be present in the roller clamp in its normal position with a 1/2 inch length extending from each end of the roller clamp. The clamp shall be held vertically, with the large end down, over a supporting gap so that the roller if forced out of the body is free to fall. A polished steel rod is applied to the top of the roller, so that the roller shall be able to rotate in the course of the test, and the vertical downward force required to separate the roller from the clamp body is measured. As ar alternate, any position of the clamp is acceptable providing the clamp body is not distorted and the load measurement is accurate.

Safety (non-toxicity) and freedom from pyrogens The extension tube shall be nontoxic (safe) and pyrogen-free when tested in accordance with the USP for Transfusion and Infusion Assemblies.

Sterility. The extension tube shall be sterile in accordance with the USP or AAMI requirements. Sterility assurance level shall be 0.000001.

Instructions. Instruction for proper use of the sets shall be provided with each unit package.

Certificates of sterility, freedom from pyrogens, and safety tests. A certificate listing all lot or control numbers shall state that representative samples from each lot or control have been tested and found to be sterile, free from pyrogens, and safe. This certificate shall include the specific tests performed and the test results. For source inspection, certificates shall be furnished to the procuring activity via the government inspector. For destination inspection, certificates shall be submitted by the contractor directly to Defense Personnel Support Center, directorate of Medical Materiel, Technical Operations Division, 2800 South 20th Street, Philadelphia, PA 19102, ATTN: Quality Assurance Branch, and one copy of each shall be forwarded with each shipment to the consignee.

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Manufacturer's records of tests. The manufacturer shall place a code or lot number on the paper label, or on the printed carton, which can be verified against the manufacturer's records of all tests made on nontoxic, sterile, pyrogen-free sets. These records shall be made available for inspection by any duly authorized Government representative. No shipment shall be made until all tests have been completed, verified, and recorded in the permanent records of the manufacturer.

Individual container. Each extension tube shall be coiled in such a manner as to prevent kinking. The coiled tube shall be packaged sterile in a sealed peel-open type container capable of maintaining sterility of contents. As an alternate, the coiled tubes shall be supplied in a paperboard box wrapped with polyester capable of maintaining sterility of contents.

Workmanship. The sets shall be free from defects which detract from their appearance or impair their serviceability.

Unit.

For Size 1 - Package (PG). One package containing twenty individually packaged sterile extension tubes, as specified, constitutes one unit. Each tube shall be individually packaged in a suitable container capable of maintaining sterility of contents unless package is damaged or opened.

For Size 2 - Each (EA). One individually packaged sterile extension tube, as specified, constitutes one unit. The tube shall be individually packaged in a suitable sealed container capable of maintaining sterility of contents unless package is damaged or opened.

For Size 3 - Package (PG). One package containing either 48 or 50 individually packaged sterile extension tubes, as specified, constitutes one unit. Each tube shall be individually packaged in a suitable sealed container capable of maintaining sterility of contents unless package is damaged or opened.

For Size 4 - Each (EA). One individually packaged sterile extension tube, as specified, constitutes one unit. The tube shall be individually packaged in a suitable sealed container capable of maintaining sterility of contents unless package is damaged or opened.

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QUALITY ASSURANCE PROVISIONS.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and as specified herein. Unit of product for examination purposes shall be one extension tube. Sampling shall be Inspection level II with an acceptable quality level (AQL) of 1.0.

Examination. Extension tubes shall be conducted to determine compliance with specification requirements.

Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and as specified herein. Unit of product for test purposes shall be one extension tube. Sampling shall be Inspection level S-2 with an AQL of 1.0.

Tests. Tests shall be conducted to determine compliance with specification requirements. Where feasible, the same sample shall be used for the determination of two or more test characteristics.

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Contractor Certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this Commercial Item Description, and that the product conforms to the producers' own drawings, specifications, standards, and quality assurance practices. The government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, providing they fall within the tolerance specified using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this document are met.

If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable.

The contracting officer has the option of accepting or rejecting the product.

Regulatory requirements.

Federal Food, Drug, and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction , the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug, and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered material. The offeror/contractor is encouraged to use recovered material in accordance with Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

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Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

Ordering data. (Intermediate/exterior package quantities and labeling and marking must be specified in the contract and/or order).

NOTE 1: The following National Stock Numbers are covered by this document:

NSN	Item Identification	Size
6515-00-782-6516	TUBE, EXTENSION, INJECTION SET, Plastic, Disposable, Clamp, 30-Inch, 20s (Type I).	1
6515-00-926-2153	TUBE, EXTENSION, INJECTION SET, Plastic, Disposable, Clamp, 10-Ft. (Type II).	2
6515-01-247-8925	TUBE, EXTENSION, INJECTION SET, Plastic, Disposable, Two Clamps, Two Y sites, 30-Inch, 48s, (Type III).	3
6515-00-959-8116	TUBE, EXTENSION, INJECTION SET, STERILE, Plastic, Disposable, Two Clamps, Two Y Sites, 30-Inch, (Type III).	4
6515-00-766-0775	TUBE, EXTENSION, INJECTION SET, STERILE, Plastic, Disposable, 20 Inch (Type IV).	4
6515-00-766-0776	TUBE, EXTENSION, INJECTION SET, STERILE, Plastic, Disposable, 30 Inch (Type V).	4

NOTE 2: Examination and testing other than sterilization testing may be waived provided certificates of compliance can be supplied assuring compliance with specification requirements as verified through process control monitoring, in-house physical testing, measuring, and visual examination.

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NOTE 3: Copies of ISO Standard 594/1 may be obtained through the American National Standards Institute (ANSI), 11 West 42nd Street, New York, NY 10036.

MILITARY INTERESTS:

Custodians:

Army-MD
Navy-MS
Air Force-03

PREPARING ACTIVITY:

DoD-MB

Agent:

DLA-DM

CIVIL AGENCY COORDINATING ACTIVITIES:

VA-OSS
USPHS
FDA-MPQAS

Project No. 6515-5492

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER: A-A-54859	2. DOCUMENT DATE (YYMMDD): 21 May 1993

3. DOCUMENT TITLE: TUBE, EXTENSION, INJECTION SET

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)
	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON
c. ADDRESS (Include Zip Code)	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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